

Product Lifecycle & Catalogue Governance

Functional Design & Backend Screens

TICVAI | Professional Reference Document

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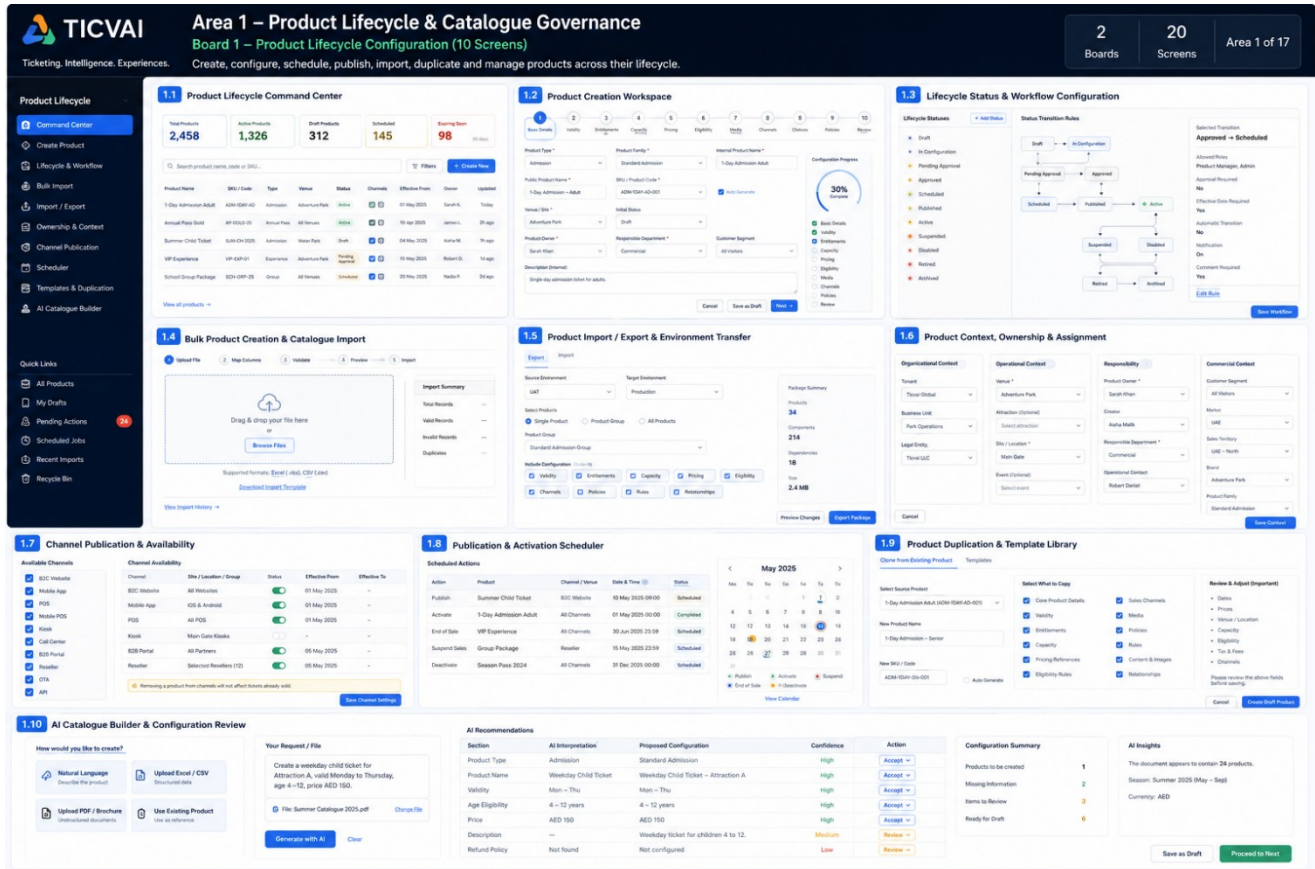
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Board 2 — Final Screen Register

Area 1 — Product Lifecycle & Catalogue Governance

Board 1 — Product Lifecycle Configuration



Board 1 establishes the operational backend workspace used by administrators and product managers to **create, configure, prepare, schedule, publish, duplicate, import/export and manage ticketing products throughout their active lifecycle**.

It covers the core Product Lifecycle requirements in the matrix, particularly product workflow/status, bulk catalogue creation, environment transfer, Draft status, publication by channel, scheduled lifecycle actions, duplication, ownership/context and AI-assisted catalogue generation.

This board deliberately does **not** cover detailed approval governance, version rollback, impact analysis, retirement/archive governance and advanced audit/change control; those belong to **Board 2 — Product Governance & Change Control**.

Screen 1 — Product Lifecycle Command Center

Purpose

Provide administrators with a centralized operational view of every product and its current lifecycle state.

Backend Functions

The screen shall:

- Display all products across venues and organizations.
- Show current lifecycle status.

- **Support configurable lifecycle statuses such as:**

- o Draft
- o In Configuration
- o Pending Approval
- o Approved
- o Scheduled
- o Published
- o Active
- o Suspended
- o Disabled
- o Retired
- o Archived

- Display product owner and responsible department.

- Show publication status by channel.

- Display effective/activation dates.

- Show pending lifecycle actions.

- **Allow advanced filtering by:**

- o Product type
- o Venue
- o Location
- o Status
- o Owner
- o Department
- o Channel
- o Effective date
- o Creation date
- o Last modification

- Provide bulk selection and authorized bulk actions.

- Flag incomplete configurations or publication blockers.

AI Capability

AI shall highlight products requiring attention, including incomplete setup, unusual configuration, approaching activation dates and potential lifecycle conflicts.

Acceptance Condition

An authorized administrator can locate any product, identify its exact lifecycle state and determine the next required operational action from one screen.

Screen 2 — Product Creation Workspace

Purpose

Provide a governed starting point for creating a new ticketing product.

Backend Functions

Administrators shall be able to:

- Create a product from scratch.
- Select product family/type.
- Assign internal product name.
- Define public-facing product name.
- Generate unique product/SKU/PLU identifier.
- Assign venue/site/location.
- Assign product owner.
- Assign responsible department.
- Select applicable customer segment.
- Define initial lifecycle status.
- Add internal description and notes.
- Add multilingual sales descriptions.
- Upload product imagery/documents.
- Select configuration template where available.
- Save incomplete configuration as Draft.
- Prevent Draft products from appearing on sales channels.
- Define required configuration sections before publication.

The workspace should show configuration completeness, for example:

Product → Validity → Entitlements → Capacity → Pricing → Eligibility → Media → Channels → Policies → Publication

These sections may be configured in their respective TICVAI engines rather than duplicating their configuration logic here.

AI Capability

The administrator can describe the desired product in natural language, for example:

“Create a weekday child ticket for Attraction A, valid Monday–Thursday, age 4–12.”

AI can prepare a proposed product configuration for administrator review.

Acceptance Condition

A new product can be created and saved in Draft without becoming commercially available until all required configuration and governance conditions are satisfied.

Screen 3 — Lifecycle Status & Workflow Configuration

Purpose

Configure how products move between lifecycle states.

Backend Functions**Administrators shall configure:**

- Available lifecycle statuses.
- Status sequence.
- Allowed status transitions.
- Which roles can initiate each transition.
- Required information before transition.

- Whether approval is required.
- Whether transition is manual or automatic.
- Notification triggers.
- Effective-date requirements.
- Validation requirements.
- Reason/comment requirements.
- Status-specific edit permissions.

Example:

Draft → Pending Approval → Approved → Scheduled → Active → Suspended → Retired → Archived

The system shall prevent unauthorized or logically invalid transitions.

Workflow Rules

Examples:

- Draft → Published cannot occur without required configuration.
- Pending Approval → Active cannot bypass required approval.
- Active → Draft may be prohibited.
- Retired products cannot automatically return to Active unless permitted by governance rules.

Detailed approval routing itself belongs to Board 2.

Acceptance Condition

Administrators can configure a controlled product lifecycle without development or database changes.

Screen 4 — Bulk Product Creation & Catalogue Import

Purpose

Allow large catalogues to be created efficiently rather than configuring every product manually.

This directly addresses the matrix requirement for catalogue creation through bulk-file upload.

Supported Sources

The system should support:

- Excel
- CSV
- Structured spreadsheet templates
- Existing catalogue files
- PDF brochures
- Product documents

Backend Functions

The administrator can:

- Download an import template.
- Upload catalogue data.

- Map source columns to TICVAI fields.
- Select target venue/site.
- Select default product configuration.
- Validate records before import.
- Preview products before creation.
- Correct errors.
- Exclude selected records.
- Run import.
- View successful/failed records.
- Download an error report.
- Roll back an import where permitted.

AI Capability

AI shall analyze semi-structured sources such as spreadsheets, brochures, PDFs and existing catalogues and propose:

- Products
- Product types
- Prices
- Validity
- Rules
- Entitlements
- Categories
- Descriptions
- Potential relationships

Nothing generated by AI should become live automatically.

Acceptance Condition

An administrator can convert a large existing catalogue into draft TICVAI products through a controlled import and validation process.

Screen 5 — Product Import / Export & Environment Transfer

Purpose

Allow controlled movement of product configurations between TICVAI environments.

This covers the matrix requirement for importing/exporting catalogue products between different environments.

Example Environments

- Development
- Sandbox
- UAT
- Staging
- Production

Backend Functions

Users can:

- Select products for export.
- Export individual products.
- Export product groups.
- Select associated configuration components.
- Create configuration packages.
- Import packages into another environment.
- Map venue/location references.
- Map dependencies.
- Detect missing references.
- Compare source and target configuration.
- Validate before deployment.
- Resolve conflicts.
- Preview changes.
- Execute transfer.
- View transfer result.

Security

Production imports may require elevated permission and/or Board 2 approval governance.

Acceptance Condition

Products can be moved between authorized environments without manually recreating their configuration and without silently overwriting incompatible production settings.

Screen 6 — Product Context, Ownership & Assignment

Purpose

Define **where the product belongs and who is responsible for it.**

Backend Functions

Configure:

Organization

- Tenant
- Business unit
- Legal entity

Operational Context

- Venue
- Attraction
- Event
- Site
- Location

Responsibility

- Product owner
- Creator
- Responsible department

- Operational contact

Commercial Context

- Customer segment
- Market
- Sales territory
- Brand
- Product family

The matrix specifically requires tracking product owners, creators, approvers and responsible departments and assigning products to locations, venues and customer segments.

Permissions

Product editing rights can inherit from organizational and venue hierarchy.

Acceptance Condition

Every governed product has clearly defined ownership, organizational context and operational/commercial assignment.

Screen 7 — Channel Publication & Availability

Purpose

Control **where a product may be exposed for sale.**

Supported Channels

For example:

- B2C Website
- Customer Mobile App
- POS
- Mobile POS
- Kiosk
- Call Center
- B2B Portal
- Reseller
- OTA
- API
- Partner marketplace

Backend Functions

Administrators can:

- Enable/disable channels.
- Select specific channels.
- Select specific sites/webstores.
- Select specific POS groups.
- Assign product availability by venue.
- Configure channel-specific effective dates.
- Preview publication status.

- Detect missing channel dependencies.
- Remove a product from sale without invalidating previously sold tickets.

Detailed channel configuration belongs to the separate **Sales Channel Management** area; this screen manages the product's association with those channels.

Important Rule

Removing a product from sale ≠ invalidating previously sold tickets.

This distinction must be visible in the UI.

Acceptance Condition

An authorized user can control precisely which approved channels expose a product while previously issued valid entitlements remain unaffected unless a separate governed action explicitly changes them.

Screen 8 — Publication & Activation Scheduler

Purpose

Automate future product lifecycle actions.

Configurable Actions

Administrators can schedule:

- Publication
- Sales start
- Activation
- Sales suspension
- Deactivation
- End of sale
- Retirement trigger

Configuration

Each scheduled action shall support:

- Date
- Time
- Time zone
- Venue
- Channel
- Product
- Status transition
- Notification
- Pre-action validation
- Failure handling

Calendar View

Provide:

- Upcoming activations
- Upcoming publications
- End-of-sale dates

- Scheduled suspensions
- Lifecycle conflicts
- Failed scheduled jobs

AI Capability

AI can identify conflicting or suspicious schedules, such as a product scheduled to become available after its configured validity period.

Acceptance Condition

Products can automatically enter or leave defined commercial lifecycle states at configured times without manual intervention.

Screen 9 — Product Duplication & Template Library

Purpose

Accelerate product configuration by allowing administrators to reuse proven configurations.

The source matrix explicitly requires duplication of products together with associated configuration, rules, pricing and entitlements.

Duplication Options

Users can select what to copy:

- Core product details
- Validity
- Pricing references
- Entitlements
- Capacity references
- Eligibility
- Sales channels
- Media
- Policies
- Rules
- Content
- Images
- Relationships

Template Library

Administrators can create reusable templates such as:

- Standard Admission
- Child Admission
- VIP Ticket
- Timeslot Ticket
- Event Ticket
- Group Product
- Annual Pass
- Add-On
- Venue-specific templates

Smart Clone

When cloning, the system should require review of sensitive fields such as:

- Dates
- Prices
- Venue
- Capacity
- Event
- Tax
- Channels

AI Capability

AI can suggest which fields should be changed in the duplicated product based on its new context.

Acceptance Condition

Users can create a new Draft product from an existing product/template while preventing accidental reuse of inappropriate dates, venues, prices or other context-sensitive values.

Screen 10 — AI Catalogue Builder & Configuration Review

Purpose

Provide TICVAI's AI-first interface for accelerating product creation and configuration.

This directly supports the matrix requirement allowing administrators to upload spreadsheets, brochures, PDFs or existing catalogues and use AI to generate product structures, pricing, rules, entitlements and configurations.

Input Methods**Administrators may:**

- Type a natural-language request.
- Upload Excel.
- Upload CSV.
- Upload PDF.
- Upload brochure.
- Upload existing catalogue.
- Select an existing product as reference.

AI Output**AI can propose:**

- Product structure
- Ticket type
- Product name/descriptions
- Validity
- Pricing structure
- Entitlements
- Eligibility
- Capacity requirements

- Channel assignments
- Media requirements
- Product relationships
- Policies
- Missing information

AI Review Workspace

Each recommendation should be classified as:

Accepted / Modified / Rejected / Requires Review

The administrator should see:

Source → AI interpretation → Proposed TICVAI configuration

AI Confidence

Where practical, show confidence or review warnings for ambiguous extracted information.

For example:

Price: AED 250 — High confidence

Child age: 4-12 — High confidence

Validity: “Summer season” — Requires clarification

Refund policy: Not found — Missing configuration

Governance

AI shall **never bypass TICVAI approval, validation or authorization rules.**

AI creates or recommends configuration; authorized users remain responsible for approving and publishing it.

Acceptance Condition

An administrator can transform unstructured or semi-structured commercial information into a structured Draft product configuration, review every AI-generated recommendation and approve or modify it before the normal lifecycle workflow begins.

Board 1 — Final Screen Register

Screen	Backend Screen
1.1	Product Lifecycle Command Center
1.2	Product Creation Workspace
1.3	Lifecycle Status & Workflow Configuration
1.4	Bulk Product Creation & Catalogue Import
1.5	Product Import / Export & Environment Transfer
1.6	Product Context, Ownership & Assignment
1.7	Channel Publication & Availability

Screen	Backend Screen
1.8	Publication & Activation Scheduler
1.9	Product Duplication & Template Library
1.10	AI Catalogue Builder & Configuration Review

Boundary with Board 2

Board 1 answers:

“How do I create, prepare, organize and publish a product?”

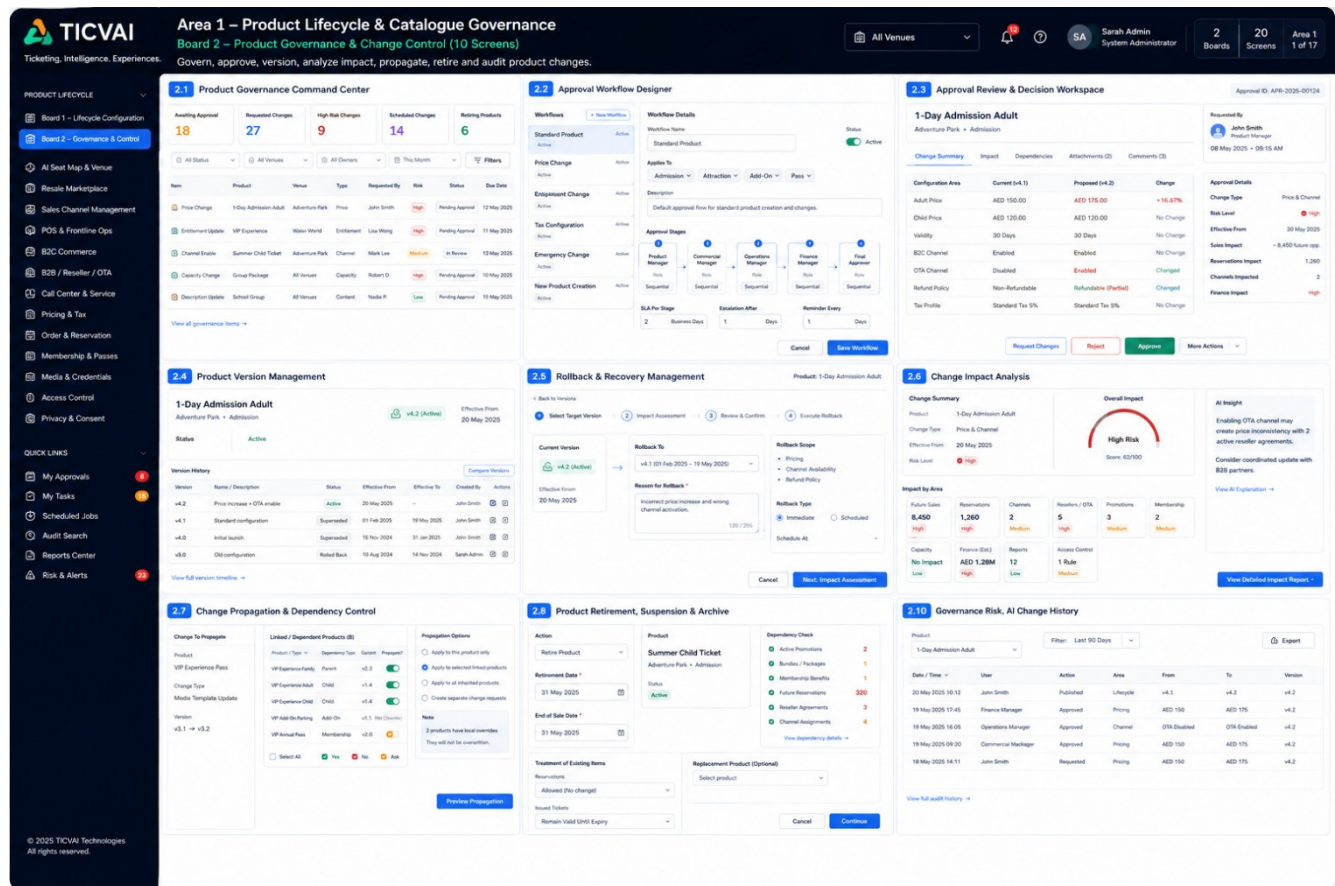
Board 2 will answer:

“How do I govern, approve, version, change, assess impact, retire and audit that product safely?”

That separation is important. It prevents approval/version/audit functionality from being repeated across multiple screens and gives us a much cleaner backend architecture for the eventual design.

Area 1 — Product Lifecycle & Catalogue Governance

Board 2 — Product Governance & Change Control



Board Objective

Board 2 governs what happens **after a product has been created or while an existing product is being changed.**

Board 1 answered:

How do we create, configure, organize and publish a product?

Board 2 answers:

How do we control changes, approvals, versions, dependencies, impact, retirement and audit without creating operational or commercial risk?

This board covers the matrix requirements for approval workflows, versioning, rollback, controlled change propagation, impact analysis, retirement/archiving, change history, scheduled lifecycle governance and ownership/accountability.

Screen 2.1 — Product Governance Command Center

Purpose

Provide management and administrators with a single control center for all product governance activities.

Backend Functions

The dashboard shall display:

- Products awaiting approval
- Changes awaiting approval
- Rejected changes
- Products with governance warnings
- Scheduled changes
- Products with unpublished changes
- Products with dependency conflicts
- Products approaching retirement
- Recently published versions
- Failed publications/rollbacks
- High-risk configuration changes

Governance Queue

Each record should display:

- Product
- Venue
- Product owner
- Change type
- Current version
- Proposed version
- Requested by
- Requested date
- Risk level
- Approval status
- Effective date
- Impacted channels
- Assigned approver

Filters

Filter by:

- Venue
- Product type
- Owner
- Department
- Status
- Risk
- Change type
- Approver
- Effective date
- Channel

AI Capability

AI shall prioritize governance tasks based on potential business impact.

Example:

High Risk — Price and entitlement modified for an active product with 8,450 future bookings.

Acceptance Condition

Authorized users can identify every product requiring governance attention and prioritize actions from a single dashboard.

Screen 2.2 — Approval Workflow Designer

Purpose

Configure reusable approval workflows governing product creation and modification.

Backend Functions

Administrators shall be able to build approval workflows using configurable stages.

Example:

Product Manager → Commercial Manager → Operations → Finance → Final Approval

Configuration

Each workflow can define:

- Workflow name
- Applicable product types
- Venue
- Department
- Change type
- Approval stages
- Approver role
- Specific approver
- Approval group
- Sequential/parallel approval
- Mandatory/optional stage
- SLA
- Escalation
- Delegation
- Reminder frequency
- Rejection behavior
- Resubmission behavior

Conditional Approval

Different changes can trigger different workflows.

For example:

Description change

→ Product Manager only

Price change

→ Commercial + Finance

Capacity change

→ Operations

Entitlement change

→ Operations + Access Control

Tax configuration

→ Finance

Acceptance Condition

Administrators can configure product approval workflows without development work and route different types of changes through different approval paths.

Screen 2.3 — Approval Review & Decision Workspace

Purpose

Give approvers a clear interface for reviewing a proposed product or product change before making a decision.

Change Comparison**Display:**

Configurati on	Current	Propose d
Adult Price	AED 150	AED 175
Validity	30 days	60 days
B2C	Enabled	Enabled
OTA	Disabled	Enabled
Refund	Non- refundable	Refundab le

Changes should be visually highlighted.

Approver Actions

- Approve
- Reject
- Request Changes
- Add Comment
- Reassign
- Delegate
- Escalate
- View Impact
- View Dependencies
- View History

Supporting Information**Approvers can see:**

- Requester
- Reason for change
- Business justification

- Attachments
- Effective date
- Current sales
- Future reservations
- Channels affected
- Pricing impact
- Capacity impact
- Finance impact
- Access impact

AI Capability

AI can summarize the proposed change:

“This request increases the adult ticket price from AED 150 to AED 175 and enables OTA distribution. Approximately 4,200 future selling opportunities may be affected.”

AI does **not** approve changes.

Acceptance Condition

An approver can understand exactly what is changing, why it is changing and what it may affect before approving or rejecting the request.

Screen 2.4 — Product Version Management

Purpose

Maintain controlled versions of every governed product configuration.

The source matrix specifically requires version history for products and the ability to roll back to earlier versions.

Version Information

Each version shall record:

- Version number
- Version name
- Product
- Created by
- Creation date/time
- Approved by
- Approval date
- Effective from
- Effective to
- Status
- Change reason
- Related approval request
- Publication status

Version States

For example:

- Draft Version

- Pending Approval
- Approved
- Scheduled
- Current
- Superseded
- Rolled Back

Comparison

Users can compare:

Version 4.2 ↔ Version 4.1

by configuration area:

- Basic product information
- Validity
- Pricing
- Capacity
- Entitlements
- Eligibility
- Media
- Channels
- Policies
- Relationships

Acceptance Condition

Every governed product modification creates a traceable version, and authorized users can compare any two versions.

Screen 2.5 — Rollback & Recovery Management

Purpose

Safely restore an earlier product configuration when a newly published configuration causes an issue.

Backend Functions

Authorized users can:

- Select current version.
- Select previous version.
- Compare versions.
- Identify dependencies.
- Run impact assessment.
- Select rollback scope.
- Enter rollback reason.
- Request emergency approval if required.
- Schedule rollback.
- Execute immediate rollback where authorized.
- Monitor rollback status.

Important Rule

Rollback should **not blindly reverse transactional history**.

For example, if Version 5 changed a price and tickets were sold under Version 5, reverting to Version 4 should not rewrite those historical transactions.

Rollback Scope**Potentially:**

- Entire product
- Pricing association
- Channel association
- Validity configuration
- Media
- Policy
- Entitlement configuration

subject to system governance.

Emergency Governance

High-authority users may have an emergency rollback process with mandatory reason and enhanced audit logging.

Acceptance Condition

An authorized user can restore a safe earlier configuration without corrupting historical sales, reservations or issued entitlements.

Screen 2.6 — Change Impact Analysis

Purpose

Show administrators **what will be affected before a product change is approved or published**.

This directly addresses the matrix requirement to perform impact analysis before product changes are published.

Impact Areas**Analyze effects on:**

- Future orders
- Reservations
- Issued tickets
- Capacity
- Pricing
- Tax
- Promotions
- Membership
- Entitlements
- Access control
- Sales channels
- B2B partners

- OTAs
- POS
- B2C
- Kiosk
- Media
- Finance
- Reporting

Example

Proposed Change:

Change ticket validity from:

90 days → 30 days

Impact analysis:

- 12,840 future ticket sales opportunities
- 1,260 existing reservations
- 5 reseller agreements
- 3 promotions
- 2 membership benefits
- 7 channels

Risk Classification

- Low
- Medium
- High
- Critical

AI Capability

AI shall explain impact in business language and identify unusual consequences.

Example:

“This change may create inconsistent validity between direct B2C sales and two reseller allocations.”

Acceptance Condition

Before a material configuration change is approved or published, authorized users can see its potential cross-module and transactional impact.

Screen 2.7 — Change Propagation & Dependency Control

Purpose

Control whether approved product changes should automatically propagate to related products or dependent configurations.

The source matrix requires controlled propagation of changes to linked products.

Dependency Types

A product may be linked to:

- Parent product
- Child product

- Bundle
- Add-on
- Upgrade
- Membership
- Package
- Promotion
- Price profile
- Capacity pool
- Entitlement
- Sales channel
- Media template

Propagation Options

For each change:

Apply to this product only

Apply to selected linked products

Apply to all inherited products

Create separate change requests

Example

Master product:

General Admission

Linked products:

- Adult Admission
- Child Admission
- Senior Admission
- Resident Admission

If the master media template changes, administrators may choose whether the change propagates.

Conflict Detection

System should detect when a dependent product has a local override that would be overwritten.

Acceptance Condition

Product changes can propagate through defined relationships without unintentionally overwriting approved local exceptions.

Screen 2.8 — Product Retirement, Suspension & Archive

Purpose

Provide a governed end-of-life process for products.

The source matrix explicitly requires disabling or retiring products without affecting previously sold tickets.

Available Actions

- Suspend Sales

- Temporarily Disable
- End Sale
- Retire
- Archive

Retirement Configuration

Administrator defines:

- Retirement date
- End-of-sale date
- Channels
- Reason
- Replacement product
- Existing reservation treatment
- Existing ticket treatment
- Communication requirements
- Reporting treatment

Critical Principle

The system must distinguish:

Product sales availability

from

validity of previously issued tickets/entitlements.

Retiring a product should not automatically invalidate previously sold valid tickets.

Dependency Check

Before retirement, show:

- Active promotions
- Bundles
- Membership benefits
- Reseller agreements
- Future reservations
- Active price lists
- Channel assignments
- Upgrade paths

Acceptance Condition

A product can be safely removed from future sale while protecting existing valid customer transactions and identifying unresolved dependencies.

Screen 2.9 — Product Audit Trail & Change History

Purpose

Provide a complete, immutable history of product configuration and governance activity.

The matrix requires tracking configuration changes with timestamps and user information.

Audit Record

Each entry should capture:

- Date/time
- User
- Role
- Product
- Version
- Action
- Configuration area
- Previous value
- New value
- Reason
- Approval reference
- Source channel
- Environment
- IP/device metadata where applicable

Search & Filters**Filter by:**

- Product
- User
- Date
- Action
- Version
- Configuration area
- Venue
- Approval
- Risk
- Environment

Timeline View**Example:**

09:05 — Sarah created Version 4.2

09:18 — Price changed AED 150 → AED 175

10:03 — Finance approved

10:12 — Commercial approved

11:00 — Version 4.2 published

Export

Authorized users can export audit reports for compliance or investigation.

Acceptance Condition

Every material product configuration and governance action is traceable to who performed it, when it occurred, what changed and under which approval.

Screen 2.10 — Governance Risk, AI Monitoring & Control Center

Purpose

Use TICVAI intelligence to continuously identify catalogue governance risks rather than relying entirely on administrators to discover them manually.

AI Monitoring

AI should monitor for:

- Products without owners
- Products without required approval
- Products with outdated pricing
- Conflicting validity
- Missing channel configuration
- Orphaned dependencies
- Unused products
- Duplicate products
- Unusual configuration changes
- Products with unusually high override levels
- Scheduled publication conflicts
- Expired commercial configuration
- Products still active after an event ends
- Products with broken dependencies

Risk Dashboard

Display:

Critical / High / Medium / Low

with:

- Risk
- Product
- Venue
- Business impact
- Recommended action
- Owner
- Due date
- Status

AI Recommendations

Example:

Potential duplicate detected

“Summer Child Admission” and “Child Summer Admission” share 96% of their configuration.

Or:

Governance warning

Product remains active on B2B although its B2C end-of-sale date passed 12 days ago.

Human Control

AI may:

- Detect
- Explain
- Recommend
- Prioritize
- Draft remediation actions

AI shall **not independently approve, publish, retire or materially modify governed products** unless TICVAI's configured authorization model explicitly permits an automated action.

Acceptance Condition

Administrators receive proactive governance intelligence with clear explanations and recommended actions while material product decisions remain controlled by the configured authorization framework.

Board 2 — Final Screen Register

Screen	Backend Screen	Primary Responsibility
2.1	Product Governance Command Center	Governance oversight
2.2	Approval Workflow Designer	Approval configuration
2.3	Approval Review & Decision Workspace	Approve/reject changes
2.4	Product Version Management	Version control
2.5	Rollback & Recovery Management	Restore previous configurations
2.6	Change Impact Analysis	Pre-change impact assessment
2.7	Change Propagation & Dependency Control	Linked-product governance
2.8	Product Retirement, Suspension & Archive	End-of-life governance
2.9	Product Audit Trail & Change History	Complete traceability
2.10	Governance Risk, AI Monitoring & Control Center	AI governance intelligence